4.4. Prove that the following formula is a tautology.

$$U_4 = \underbrace{(p \wedge q \to r)}_{A} \to \underbrace{(p \to (q \to r))}_{B}$$

Theoretical results

1) U is a tautology ($\models U$) if $(\forall) i, i(U) = T$)

2) U is consistent if $\exists i, i(U) = T$, i-model.

3) U is inconsistent if $(\forall) i, i(U) = F$.

4) U is contingent if $\exists i, i(U) = T$, $i \to$ model of U. $\exists j, j(U) = F$, $j \to$ anti-model of U.

	p	q	r	$p \wedge q$	A	$q \to r$	B	U_4
i_1	T	T	T	T	T	T	T	T
i_2	T	T	F	T	F	F	F	T
i_3	T	F	T	F	T	T	T	T
i_4	T	F	F	F	T	F	T	T
i_5	F	T	T	F	T	T	T	T
i_6	F	T	F	F	T	F	T	T
i_7	F	F	T	F	T	T	T	T
i_8	F	F	F	F	T	F	T	T

Conclusion: for $(\forall) i_n$, $i(U) = T$, so $U \to$ valid formula, tautology.

2. Using the truth ... decide the type of the formula.

$$U_1 = \underbrace{q \vee \neg p \vee r}_{A} \to \underbrace{\neg(p \vee r)}_{B}$$

	p	q	r	$\neg p$	$p \vee r$	B	A	U_1
i_1	T	T	T	F	T	F	T	F
i_2	T	T	F	F	T	F	T	F
i_3	T	F	T	F	T	F	T	F
i_4	T	F	F	F	T	F	F	T
i_5	F	T	T	T	T	F	T	F
i_6	F	T	F	T	F	T	T	T
i_7	F	F	T	T	T	F	T	F
i_8	F	F	F	T	T	T	T	T

$\Rightarrow U_1$ contingent formula having 3 models (i_4, i_6, i_8)

model $i_4 : \{p, q, r\} \to \{T, F\}$

$i_4(p) = T$, $i_4(q) = F$, $i_4(r) = T$, $i_4(U_1) = T$.

anti-model i_1:

$i_1 : \{p, q, r\} \to \{T, F\}$

$i_1(p) = T$, $i_1(q) = T$, $i_1(r) = T$; $i_1(U_1) = F$.

Ex. 5.4 Find DNF/CNF ; valid formulas?

$$U_4 = (p \xrightarrow{1} (q \vee r)) \xrightarrow{2} ((p \xrightarrow{3} q) \vee (p \xrightarrow{4} r))$$

DNF : $\vee(\wedge \ell_{ij})$

CNF : $\wedge(\vee \ell_{ij})$?

$p \vee \neg p \vee q \equiv T$

$p \wedge \neg p \wedge \neg r \equiv F$

$A \to B \equiv \neg A \vee B$

1) Replace the „→".

$$\overset{1,3}{\equiv} \neg p \vee (q \vee r) \rightarrow (\neg p \vee q \vee \neg p \vee r)$$

$$\overset{2}{\equiv} \neg(\neg p \vee q \vee r) \vee (\neg p \vee q \vee \neg p \vee r)$$

$$\overset{\text{De Morgan}}{\equiv} (p \wedge \neg q \wedge \neg r) \vee (\neg p \vee q \vee \neg p \vee r)$$

$$\equiv (p \wedge \neg q \wedge \neg r) \vee (\neg p \vee q \vee r)$$

$A . A \equiv A$

$\neg p \vee \neg p = \neg p$

Idempotency law ?

=> DNF with 4 cubes.

$$\overset{\text{Distrib.}}{\equiv} p \vee (\neg p \vee q \vee r) \wedge (\neg q \vee \neg p \vee q \vee r) \wedge (\neg r \vee \neg p \vee q \vee r)$$

=> CNF with 3 clauses.

$$\equiv T \wedge T \wedge T \equiv T. \quad \Rightarrow \text{valid formula.}$$

7.1. Using the appropriate normal form, prove that the following formulas are inconsistent:

$$U_1 = (p \xrightarrow{1} (q \xrightarrow{2} r)) \wedge \neg((p \xrightarrow{3} q) \xrightarrow{4} (p \xrightarrow{5} r))$$

$$\underset{1,2,3,4,5}{\overset{\text{replace}}{\equiv}} \neg p \vee (\neg q \vee r) \wedge \neg(\neg(\neg p \vee q) \vee (\neg p \vee r))$$

$$\overset{\text{De Morgan}}{\equiv} (\neg p \vee \neg q \vee r) \wedge ((\neg p \vee q) \wedge \neg(\neg p \vee r))$$

$$\overset{\text{De Morgan}}{\equiv} (\neg p \vee \neg q \vee r) \wedge (\neg p \vee q) \wedge p \wedge \neg r \quad \Rightarrow \text{CNF with 4 clauses.}$$

$(A \vee B) \wedge A \equiv A$

$(A \wedge B) \vee A \equiv A$

} ABSORBTION LAWS

$$\overset{\text{Distrib.}}{\equiv} (\neg p \wedge \neg p \wedge p \wedge \neg r) \vee (\neg p \wedge q \wedge p \wedge \neg r) \vee (\neg q \wedge \neg p \wedge p \wedge \neg r) \vee$$
$$\vee (\neg q \wedge q \wedge p \wedge \neg r) \vee (r \wedge \neg p \wedge p \wedge \neg r) \vee (r \wedge q \wedge p \wedge \neg r) \Rightarrow \text{DNF with 6 cubes.}$$

$$\equiv F \vee F \vee F \vee F \vee F \vee F = F.$$

6.1. Determine the models of the formula using DNF.

$U_1 = (p \vee q \xrightarrow{1} r) \xrightarrow{2} (p \xrightarrow{3} r) \wedge q$

Precedence: $\neg, \wedge, \vee$

$\qquad\qquad\quad * \quad +$

replace 1,2,3 $\neg(\neg(p\vee q) \vee r) \vee (p \to r) \wedge q$

DeMorgan $(p \vee q \wedge \neg r) \vee ((\neg p \vee r) \wedge q)$

Distrib. $(p \wedge \neg r) \vee (q \wedge \neg r) \vee (\neg p \wedge q) \vee (r \wedge q)$

$\Rightarrow$ DNF with 4 cubes.

Cube $p \wedge \neg r$ provides 2 models:

↳ $i_{1,2} : \{p, q, r\} \to \{T, F\}$

$i_1(p) = T, \quad i_1(q) = T, \quad i_1(r) = F$

$i_2(p) = T, \quad i_2(q) = F, \quad i_2(r) = F$

Cube $q \wedge \neg r$ provides 2 models:

↳ $i_{3,4} : \{p, q, r\} \to \{T, F\}$

$i_3(p) = T \quad i_3(q) = T \quad i_3(r) = F$

$i_4(p) = F \quad i_4(q) = T \quad i_4(r) = F$

Cube $\neg p \wedge q$ provides 2 models:

↳ $i_{5,6} : \{p, q, r\} \to \{T, F\}$

$i_5(p) = F \quad i_5(q) = T \quad i_5(r) = F$

$i_6(p) = F \quad i_6(q) = T \quad i_6(r) = T$

Cube $r \wedge q$ provides 2 models:

↳ $i_{7,8} : \{p, q, r\} \to \{T, F\}$

$i_7(p) = F \quad i_7(q) = T \quad i_7(r) = T$

$i_8(p) = T \quad i_8(q) = T \quad i_8(r) = T$

$i_1 = i_3$

$i_4 = i_5$

$i_6 = i_7$

$\Rightarrow U$ has 5 models: i_1, i_2, i_4, i_6, i_8, $(U) = T$.

8.1 Write all the anti-models of the following using CNF.

$U_1 = (q \wedge r \to r) \to (p \to r) \wedge q$ Absorbtion law

$U_1 = (\neg p \vee r) \wedge q$

$\Rightarrow$ CNF with 2 clauses.

Clause $\neg p \vee r \equiv F$ provides 2 anti-models:

$\hookrightarrow i_{1,2} : \{p, q, r\} \to \{T, F\}$

$i_1(p) = T$, $i_1(q) = T$, $i_1(r) = F$

$i_2(p) = T$, $i_2(q) = F$, $i_2(r) = F$

Clause q provides 4 anti-models:

$\hookrightarrow i_{3,4,5,6} : \{p, q, r\} \to \{T, F\}$

$i_3(p) =$	$i_3(q) =$	$i_3(r) =$
T	F	T
T		F
F		T
F		F

$i_2 = i_4 \Rightarrow U_1$ has 5 anti-models